

# Computer Science & Information Technology

## Algorithms

DPP: 1

### Analysis Of algorithm

**Q1** Sort the functions in ascending order of asymptotic(big-O) complexity.

$$f_1(n) = n, f_2(n) = 80, f_3(n) = n^{\log n}, f_4(n) = \log \log^2 n, f_5(n) = (\log n)^{\log n}$$

- (A)  $f_2(n), f_4(n), f_1(n), f_5(n), f_3(n)$   
 (B)  $f_2(n), f_1(n), f_4(n), f_5(n), f_3(n)$   
 (C)  $f_2(n), f_1(n), f_4(n), f_3(n), f_5(n)$   
 (D)  $f_1(n), f_1(n), f_4(n), f_3(n), f_2(n)$

**Q2** Consider two function  $f(n) = 10n + 2\log n$  and  $g(n) = 5n + 2(\log n)^2$ , then which of the following is correct option?

- (A)  $f(n) = \theta(g(n))$   
 (B)  $f(n) = O(g(n))$   
 (C)  $f(n) = \omega(g(n^2))$   
 (D) None of the above

**Q3** Consider two function  $f(n) = \sqrt{n}$  and  $g(n) = n \log n + n$  then  $f(n)/g(n)$  is equivalent to how many of the following given below? \_\_\_\_\_.

- (a)  $o(n^{-1/2})$   
 (n)  $O(n^{-1/2})$   
 (c)  $\Omega(1/\log n)$   
 (d)  $\theta(n^{-1/2})$

**Q4** Consider the following C-code

```
void foo (int x)
{
    int a = 1;
    if (n == 1)
        return;
    for (a=1; a ≤ n; a++)
    {
        printf("GATEWALLAH");
```

```
        break;
```

```
    }
```

```
}
```

What is the worst time complexity of above program?

- (A)  $O(1)$  (B)  $O(n)$   
 (C)  $O(\log n)$  (D)  $O(\sqrt{n})$

**Q5** Consider the following asymptotic functions :

$$f_1 = 2^n$$

$$f_2 = 1.001^n$$

$$f_3 = e^n$$

$$f_4 = n!$$

Which of the following is correct increasing order of above functions?

- (A)  $f_3, f_4, f_1, f_2$  (B)  $f_2, f_4, f_1, f_3$   
 (C)  $f_3, f_2, f_1, f_4$  (D)  $f_2, f_1, f_3, f_4$

**Q6** Consider the following functions

$$f_1(n) = 4^{2^n}$$

$$f_2(n) = n!$$

$$f_3(n) = 4^{e^n}$$

$$f_4(n) = n^{nn}$$

Which of the following is/are correct?

- (A)  $f_1(n) = O(f_2(n))$   
 (B)  $f_1(n) = O(f_4(n))$   
 (C)  $f_1(n) = O(f_3(n))$   
 (D)  $f_2(n) = O(f_3(n))$

**Q7** Consider two function  $f_1(n) = n^{2^n}$  and  $f_2(n) = n^{n^2}$  then which of the following is true.

- (A)  $f_1(n) = O(f_2(n))$   
 (B)  $f_1(n) = \theta(f_2(n))$   
 (C)  $f_1(n) = \omega(f_2(n))$



(D) None of these

**Q8**  $f(n) = \sum_{i=1}^n i^3 = x$ , choices for  $x$

- I.  $\theta(n^4)$       II.  $\theta(n^5)$   
III.  $O(n^5)$       IV.  $\Omega(n^3)$

(A) I, II, III

(B) II, III, IV

(C) I, II, III, IV

(D) I, III, IV



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## Answer Key

Q1 (A)

Q2 (A, B)

Q3 2

Q4 (A)

Q5 (D)

Q6 (B, C, D)

Q7 (C)

Q8 (D)



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## Hints & Solutions

### Q1 Text Solution:

$$80 < n$$

$$\log \log^2 n < n$$

$$\text{put } n = 10^{100}$$

$$\log(\log n)^2 = 10^{100}$$

$$\log(100)^2 < 10^{100}$$

$$4 < 10^{100}$$

$$n < n^{\log n}$$

taking log on both side

$$\log n < \log n \log n$$

$$\text{we know that } (\log n)^2 > \log n$$

$$\text{now, } (\log n)^{\log n} < n^{\log n}$$

as we can see that log n in LHS and n on RHS.

$$n < (\log n)^{\log n}$$

taking log on both sides

$$\log n < \log n * \log \log n$$

From above we conclude that growth of  $\log^* \log n$  is higher than 1.

∴ option (a) is correct.

### Q2 Text Solution:

$$f(n) = 10n + 2 \log n = O(n)$$

$$g(n) = 5n + 2(\log n)^2 = O(n)$$

$$f(n) = O(g(n))$$

$$g(n) = O(f(n))$$

$$f(n) = g(n)$$

As we know that if  $\theta$  is possible then  $O$ ,  $\Omega$  is possible.

Hence option a, b are correct.

### Q3 Text Solution:

$$f(n) = \sqrt{n}$$

$$g(n) = n \log n + n$$

$$\frac{f(n)}{g(n)} = \frac{\sqrt{n}}{n(1+\log n)} = \frac{1}{\sqrt{n}(1+\log n)} = \frac{1}{\sqrt{n} + \sqrt{n} \log n}$$

#### Option (a)

$$f(n) < c.g(n) - o$$

$$f(n) \leq c.g(n) - O$$

$$O\left(n^{-\frac{1}{2}}\right) = \frac{1}{\sqrt{n}}$$

$$f(n) \leq g(n)$$

$$\frac{1}{\sqrt{n} + \sqrt{n} \log n} = o\left(\frac{1}{\sqrt{n}}\right) \quad \text{True}$$

#### Option (b)

- If  $o$  possible then  $O$  also possible, but vice versa is not possible.
  - If  $w$  is possible then  $\Omega$  also possible, but vice versa is not possible.
- ∴ Option i, ii are true.

Hence (2) is the correct answer.

### Q4 Text Solution:

If we see carefully, loop will execute only one time because of break statement, therefore time complexity will be  $O(1)$

### Q5 Text Solution:

By observing options and given options, we can conclude that-

$$f_4 > f_3 > f_1 > f_2 \text{ increasing order is } f_2, f_1, f_3, f_4.$$

∴ Option (d) is correct.

### Q6 Text Solution:

The increasing order of given functions is  $f_2, f_2, f_3, f_4$ .

Therefore (b,c,d) are correct.

### Q7 Text Solution:

$$f_1(n) = n^{2^n} \text{ and } f_2(n) = n^{n^2}$$

$$n^{2^n} = n^{n^2}$$

Taking log on both side

$$2^n \log n \quad n^2 \log n$$

as we can see that

$2^n$  has more growth rate than

$n^2$  ∴ we conclude

$$f_2(n) < O(f_1(n)) \quad \text{or } f_1(n) = \omega(f_2(n))$$

∴ (c) is correct.



**Q8 Text Solution:**

$$\begin{aligned}\sum_{i=1}^n &= \left[ \frac{n(n+1)}{2} \right]^2 \\ &= \frac{n^2(n+1)^2}{4} \\ &= \frac{n^2[n^2+2n+1]}{4} \\ f(n) &= \frac{n^4+2n^3+n^2}{4} \\ O(n^4) \quad \Omega(n^4)\end{aligned}$$

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